Code

Survival Model Selection and Validation for HEOR in R

An HTA-aligned workflow for fitting, diagnosing, comparing, and justifying parameteric survival models

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Introduction

In health economic evaluations such as cost-effectiveness analyses, estimating the long-term health benefits of innovative treatments over a lifetime horizon is essential. However, randomized controlled trials (RCTs) typically have short follow-up periods. Therefore, following health technology assessment (HTA) guidelines, it is important to extrapolate survival data in a statistically robust and clinically justified manner to predict life expectancy for both new treatments and standard care.

This R notebook provides a reusable, HTA-oriented framework showing analysts not only how to fit survival models in R, but how to evaluate, compare, interpret, and defend the model selected for economic evaluation.

Defensible workflow and R tools

NICE-recommended survival extrapolation workflow

NICE TSD14 recommends a structured four-step procedure to ensure that survival extrapolations are both statistically sound and clinically credible.

Each step builds on the last, guiding the analyst from exploratory data inspection through to a fully justified model selection that can withstand scrutiny in a health technology appraisal.

Step 1 — Assess model assumptions & hazard structure

Examine the observed data with graphical and statistical tools to understand which survival and hazard shapes are plausible before selecting a model.

Step 2 — Fit candidate models

Fit the full standard parametric family (Exponential, Weibull, Gompertz, Log-normal, Log-logistic, Generalized Gamma) and flexible alternatives where justified.

Step 3 — Evaluate statistical fit & graphical diagnostics

Compare fitted curves against the Kaplan-Meier estimate using AIC/BIC and residual diagnostics. Visually inspect hazard plots, probability plots, and residual plots to detect systematic departures.

Step 4 — Evaluate external validity & decision uncertainty

Assess whether the extrapolated curves are clinically plausible, consistent with external evidence, and robust to alternative model choices.

R packages

Three credible specialized R packages were used to implement the workflow,

- `survival`: Fit Kaplan-Meier curves and visualizations (eg., cumulative hazard plot)
- `flexsurv`: Fit parametric and flexible survival models, residual diagnosis, and visualizations
- `GofCens`: Assess goodness-of-fit and visualizations

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Input data

The input data, censored time-to-event data (locoregional control), were digitized from an RCT study by Bonner et al, 2006, which investigated the effect of cetuximab treatment on patients with head and neck cancer.

The `IPD_ctrl_gp.csv` and `IPD_treat_gp.csv` files contain time-to-event data for the control (n=213) and treatment (n=211) arms, with follow-up duration of 50 months.

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Inspect the first rows of the combined dataset. The column

- `time` represents follow-up time in months
- `status`: 1 and 0 represent event and censored, respectively
- `treat`: 1 and 0 represent treatment and control, respectively.

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```
##       time status treat
## 1  0.00274      1     0
## 2  0.00274      1     0
## 3  0.26100      1     0
## 4  0.77900      1     0
## 5  0.77900      1     0
## 6  0.77900      1     0
## 7  1.03450      0     0
## 8  1.29000      1     0
## 9  1.81000      1     0
## 10 2.07000      1     0
## 11 2.33000      0     0
## 12 2.59000      1     0
## 13 2.59000      1     0
## 14 2.85000      1     0
## 15 2.85000      1     0
```

Step 1 - Assess model assumptions & hazard structure

Use graphical and statistical tools to determine which survival and hazard shapes are plausible before selecting a model.

Step 1.1 Log cumulative hazard plot

A (log) cumulative hazard plot is recommended to answer key questions:

- Are proportional hazards reasonable?
- What hazard shape is supported?

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To assess whether hazards are constant over time (e.g., exponential distribution), plot cumulative hazard versus time. To evaluate monotonic changes in hazard (e.g., Weibull and Gompertz distributions), plot the logarithm of cumulative hazard against the logarithm of time.

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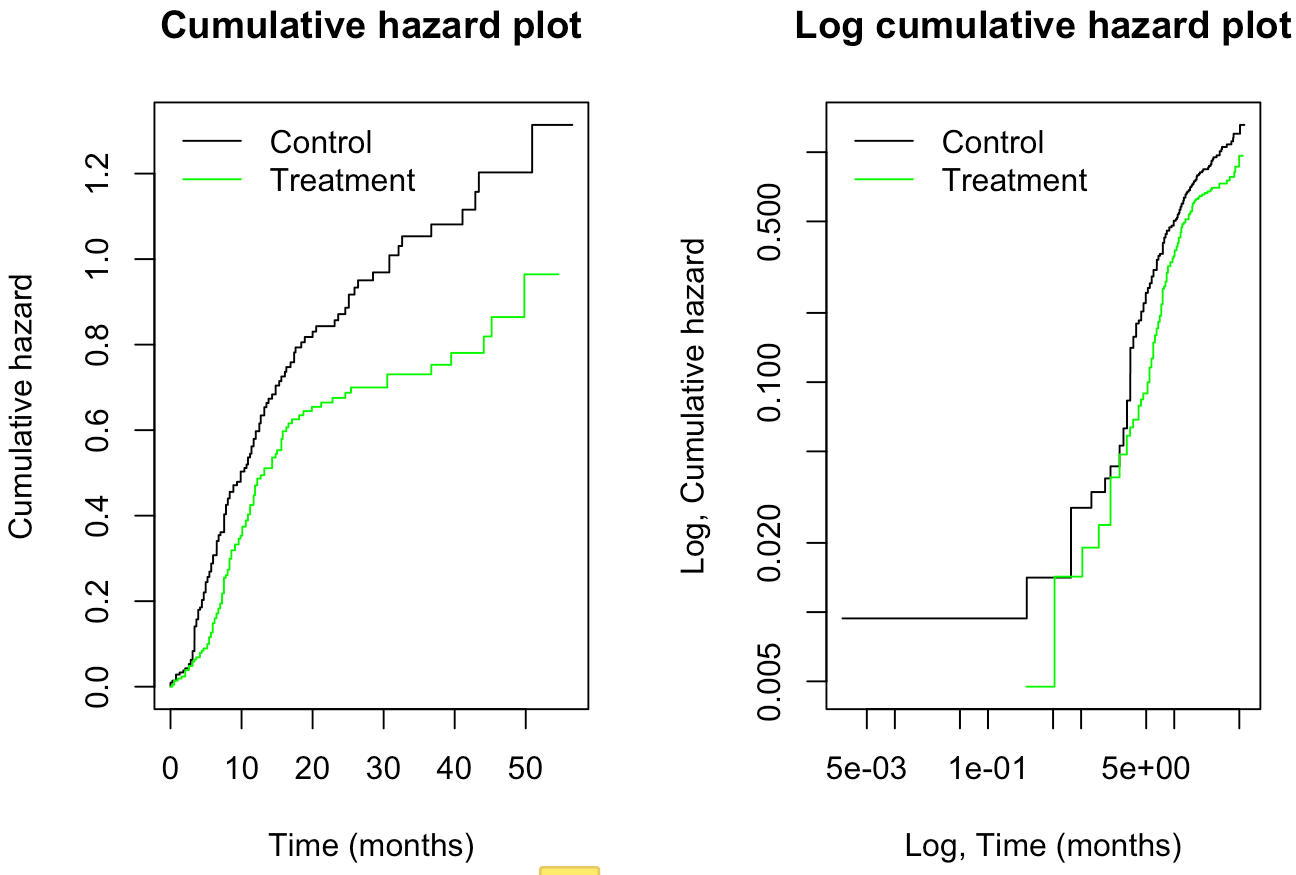


Figure 1. Log cumulative hazard plots for the control and treatment arms.

The left plot showed that cumulative hazards appeared to transition around 10-20 months, indicating the hazard is constant over the follow-up duration.

The right plot showed that cumulative hazards increased over time, suggesting the Weibull and Gompertz distribution assumptions may be reasonable.

In addition, proportional hazards are valid across the two arms in the right plot.

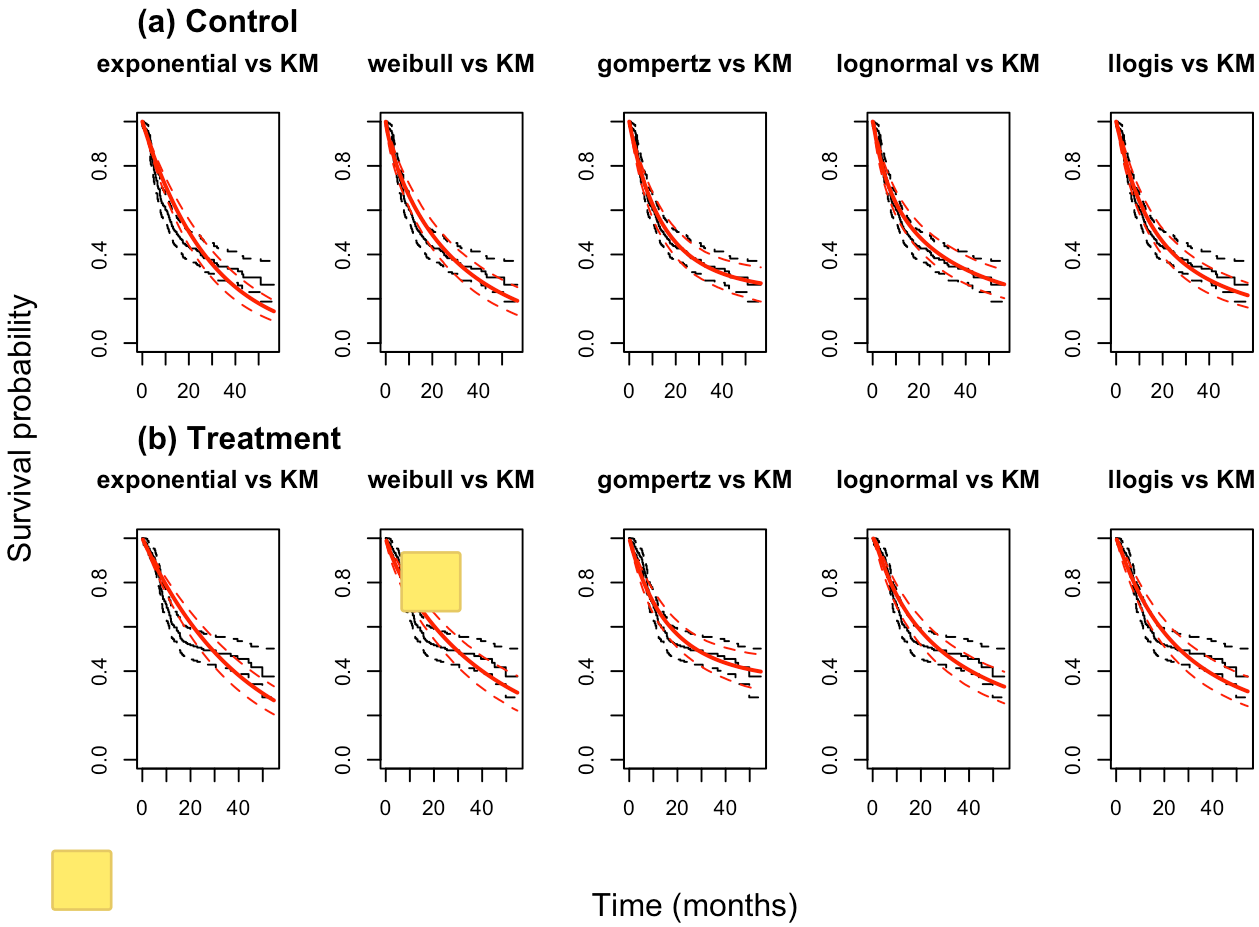
Step 1.2 Overlap KM vs parametric models

The purpose of overlapping KM curves with the chosen parametric model curves is to evaluate goodness-of-fit for various parametric survival model candidates visually.

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The package `flexsurv` can fit a variety of parametric survival models effectively. The plot shows fitted and observed data for the control and treatment arms separately.

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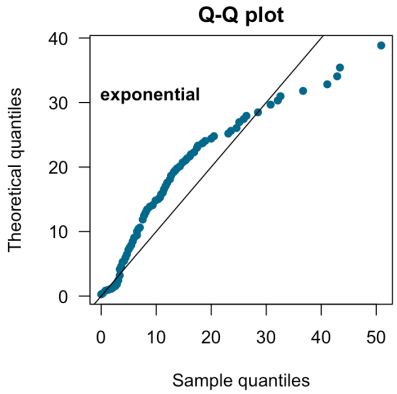


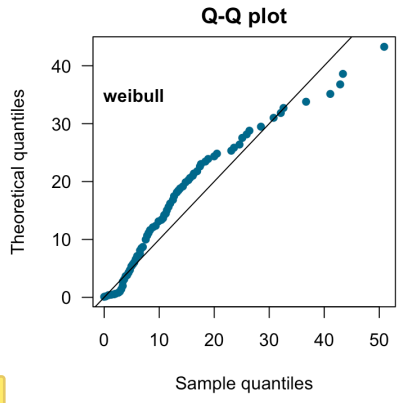
The upper and lower panels represent control and treatment arms, respectively. From visual inspection, the Gompertz, log-normal, and log-logistic models/curves appear to fit the observed trial data better than the exponential and Weibull models.

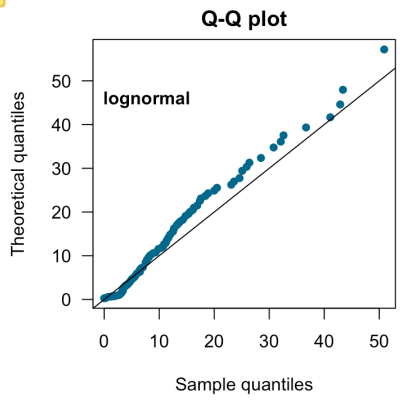
Step 1.3 Q-Q plot for right censored data

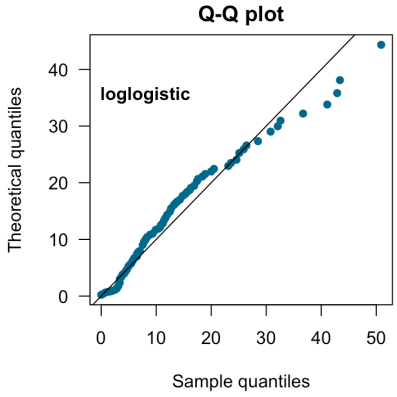
The purpose of using Q-Q plot is to determine any specific distribution families are compatible with the observed data.

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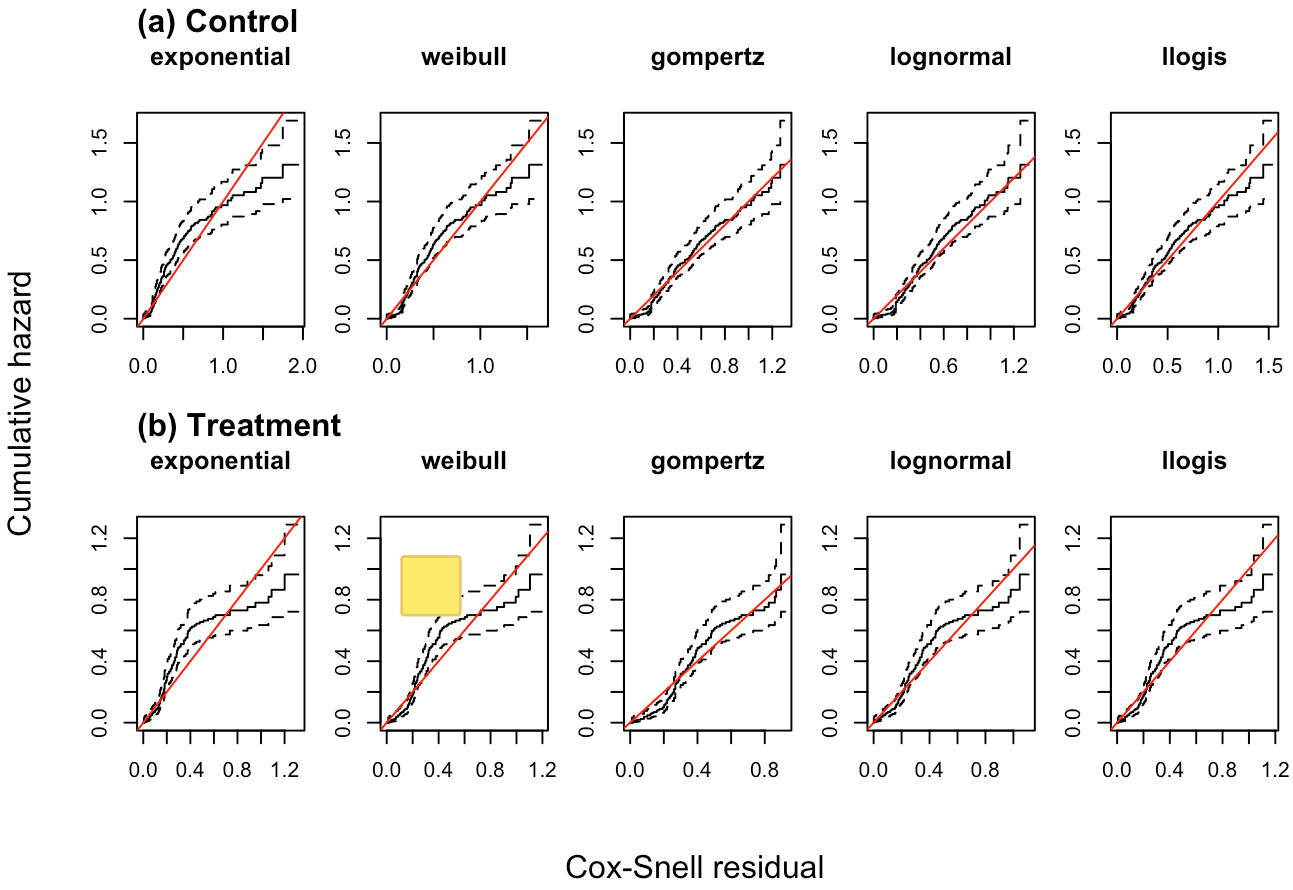


However, `GofCens` cannot support the Gompertz distribution. Moderate deviations appeared for the exponential, Weibull, log-normal, and log-logistic distributions, especially in mid-to-late follow-up.

Step 1.4 Residual plot (Cox-snell)

The residual graphics can detect any systematic lack of fit in any candidate.

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The red straight line represents a perfect goodness-of-fit between the model and the observed data. By comparing the black lines of individual models with specific distributional assumptions, we can see that Gompertz, log-normal, and log-logistic performed better for the Control arm, while Gompertz alone may have best fit the observed survival data in the Treatment arm among all model candidates.

In addition, the Cox-Snell residual is a scientifically proven measure suitable for censored survival data.

Step 2. Fit a common parametric survival models with two arms

Fit the full standard parametric family (Exponential, Weibull, Gompertz, Log-normal, Log-logistic, Generalized Gamma) and flexible alternatives where justified.

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Step 3.1 Evaluate AIC of fitted models

Evaluate robustness of the fitted model using the information criteria, AIC. The smaller the AIC scores, the more robut a model fit to the observed data.

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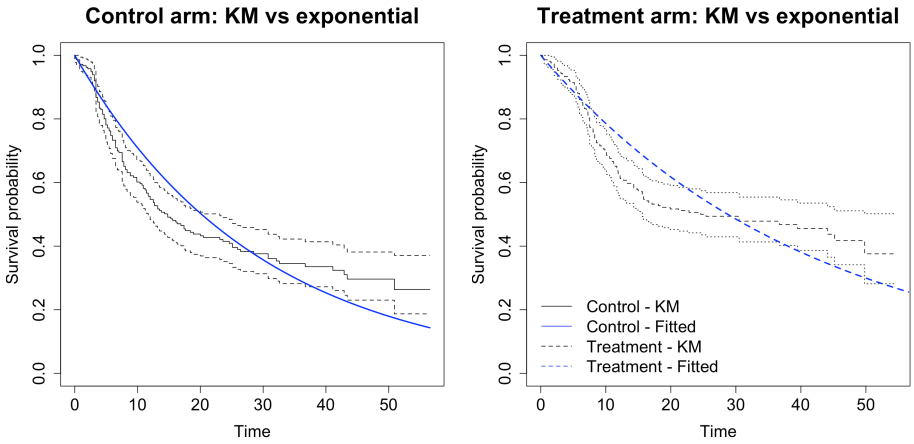
```
##   Distribution      AIC
## 3     gompertz 2173.631
## 5       llogis 2179.306
## 6     gengamma 2191.250
## 4    lognormal 2198.150
## 2      weibull 2204.574
## 1  exponential 2215.113
```

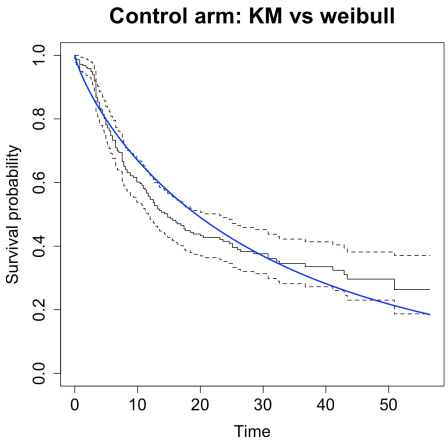
In line with the above graphics diagnostics, Gompertz model perform the best in fitting the observed trial data, among all the six model candidates.

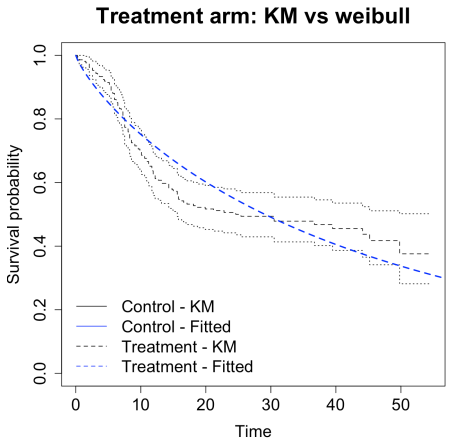
Step 3.2 Overall KM vs best fitted model

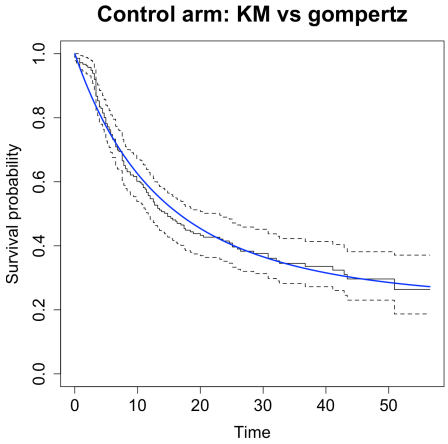
As a complemment evaluation, we visually inspect the fitted model curves, by comparing with non-parametric Kaplan-Meier curves for Control and Treatment arms, separately.

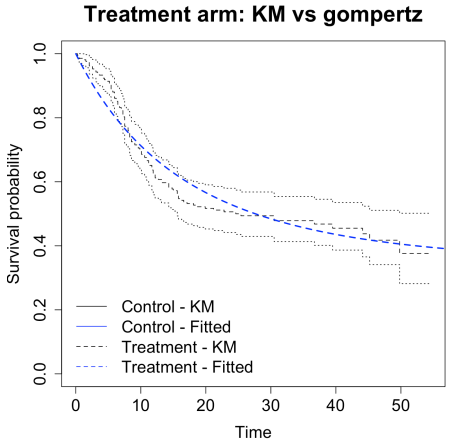
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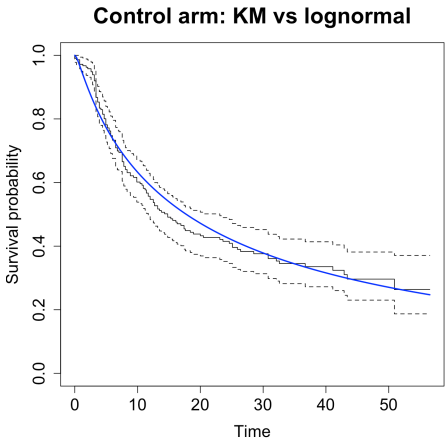


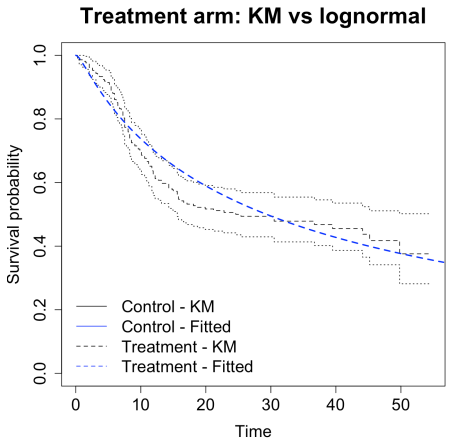

Control arm: KM vs weibull
Survival probability
Time

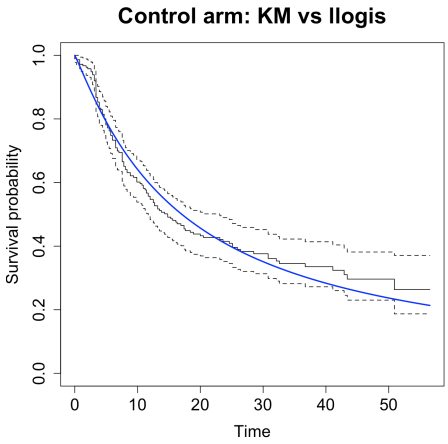

Treatment arm: KM vs weibull
Survival probability
Time
Control - KM
Control - Fitted
Treatment - KM
Treatment - Fitted

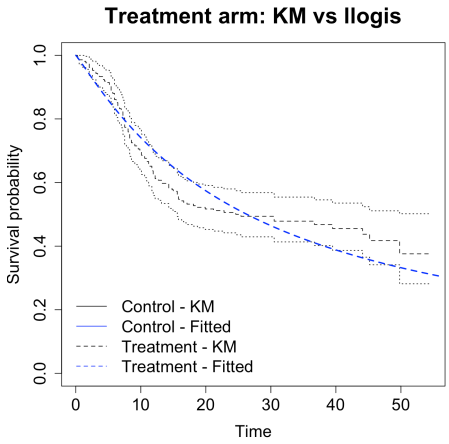

Control arm: KM vs gompertz
Survival probability
Time

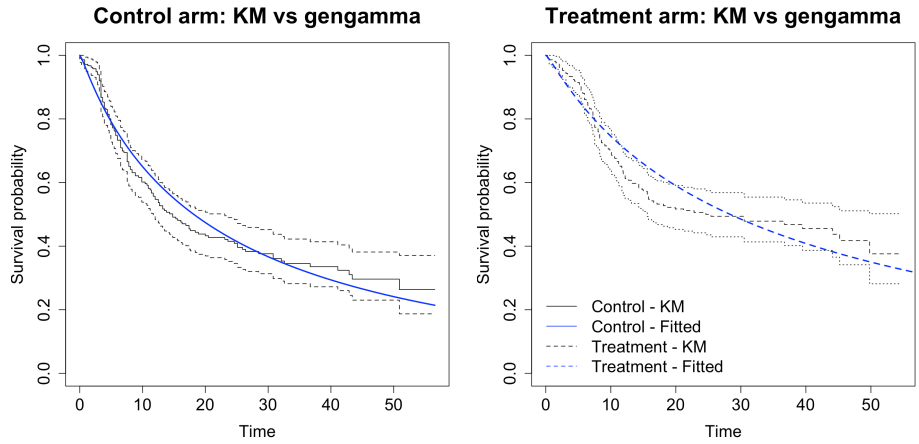

Treatment arm: KM vs gompertz
Survival probability
Time
Control - KM
Control - Fitted
Treatment - KM
Treatment - Fitted


Control arm: KM vs lognormal
Survival probability
Time


Treatment arm: KM vs lognormal
Survival probability
Time
Control - KM
Control - Fitted
Treatment - KM
Treatment - Fitted


Control arm: KM vs llogis
Survival probability
Time


Treatment arm: KM vs llogis
Survival probability
Time
Control - KM
Control - Fitted
Treatment - KM
Treatment - Fitted



In conclusion, based on the graphical fit check, the Gompertz model provides the best fit to the observed trial data.

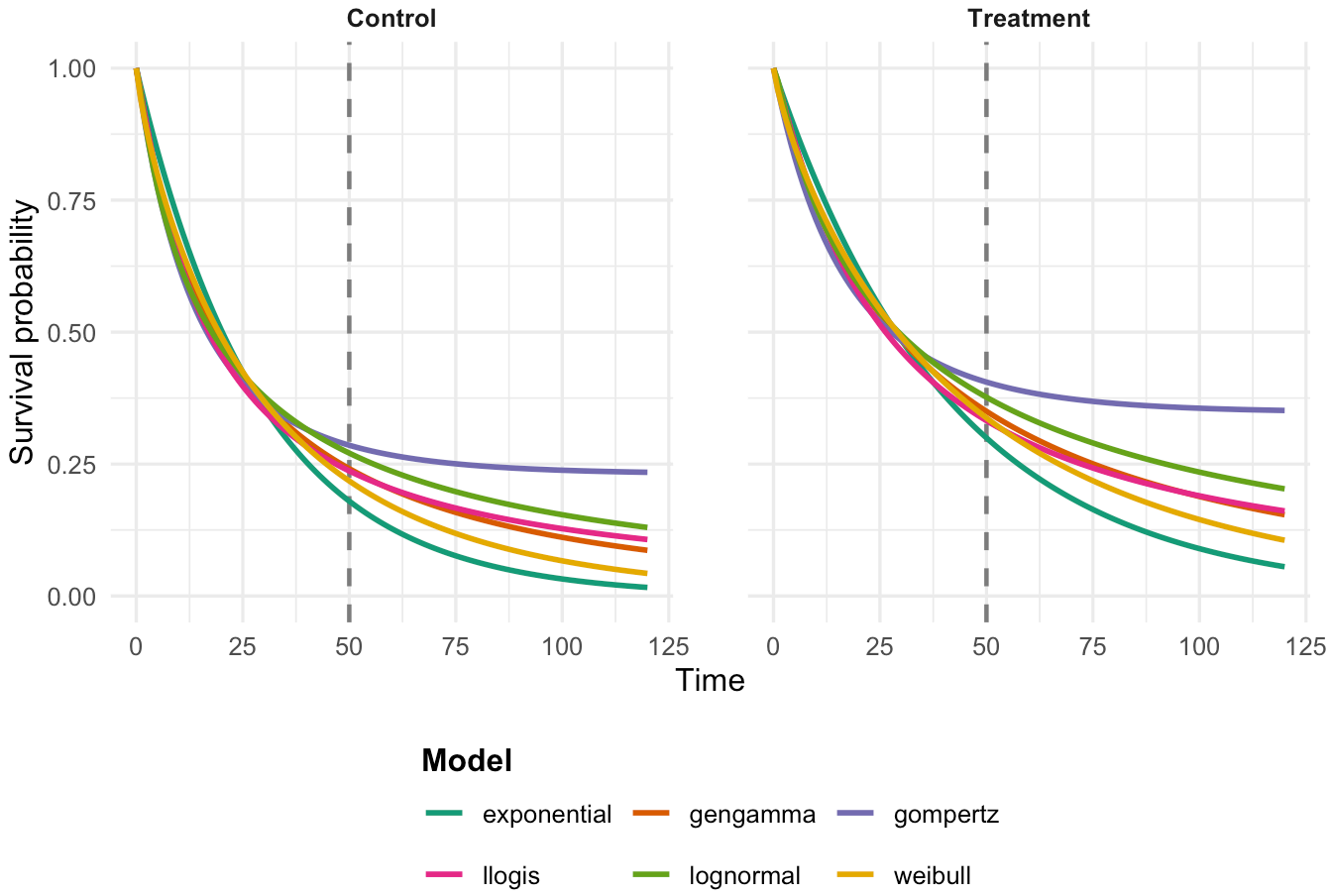
Survival extrapolation and validation

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```
## # A tibble: 10 × 6
##     time   est   lcl   ucl group   model
##    <int> <dbl> <dbl> <dbl> <chr>   <chr>
##  1     0 1     1     1     Control exponential
##  2     1 0.966 0.960 0.971 Control exponential
##  3     2 0.934 0.922 0.944 Control exponential
##  4     3 0.902 0.885 0.916 Control exponential
##  5     4 0.872 0.850 0.890 Control exponential
##  6     5 0.842 0.816 0.865 Control exponential
##  7     6 0.814 0.783 0.840 Control exponential
##  8     7 0.786 0.752 0.816 Control exponential
##  9     8 0.760 0.722 0.792 Control exponential
## 10     9 0.734 0.693 0.770 Control exponential
```

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Model Selection & Validation Summary

Author Bio

Xin (David) Zhao, PhD, integrates epidemiology, evidence synthesis, quantitative modeling, and health economics to develop transparent, reproducible analytics that support healthcare and biotech decision-making.